Systematic Literature Review Of Hate Speech Detection With Text Mining

Rini
Magister of Informatics Engineering
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
rini.26@students.amikom.ac.id

Ema Utami
Magister of Informatics Engineering
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
ema.u@amikom.ac.id

Anggit Dwi Hartanto
Faculty of Computer Science
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
anggit@amikom.ac.id

2020 2nd International Conference on Cybernetics and Intelligent System (ICORIS) | 978-1-7281-7257-6/20/$31.00 ©2020 IEEE | DOI: 10.1109/ICORIS50180.2020.9320755

***Abstract*—Along with the increasing activity on social media, hate speech is getting out of control. Hate speech detection can be done by utilizing text mining technology. There have been many hate speech detection studies conducted. To identify and analyze research trends, data sources, methods and features used in hate speech detection, this systematic literature review was created. Until early 2020, the topics of hate speech were found, including hate speech against minorities, religion, women, the general election agenda, and politics. Sources of data that are widely used to be used as datasets come from twitter. Hate speech is not only classified into HS (hate speech) and Non-HS (non-hate speech) but can be further classified into racism, sexism, offensive, abusive, threats of violence and others. Of the 38 studies that meet inclusion and exclusion, there are 26 algorithms and 28 features that have been used to detect hate speech. However, these methods and features do not necessarily guarantee a good hate detection performance. Hate speech classification performance is also influenced by the dataset, the features chosen, the number of classes and mutually exclusive classes.**

Keywords—hate speech, classification, systematic literature review, text mining

I. INTRODUCTION

As of January 2019 Indonesia had 150 million active internet users and 150 million social media users. Social media is a digital communication media where users can easily participate and spread information. With social media, citizens can easily express their aspirations to the government. However, freedom of speech on social media has a negative impact. Head of the Public Information Bureau of the National Police Headquarters Brigjen. Pol. Dedi Prasetyo in an interview with Bisnis.com revealed that in 2018 there were about 255 cases of hate speech, and in the January-June 2019 period, there were 101 cases of hate speech crimes handled by the National Police.

A number of prohibitions on hate speech in the form of statutory regulations have been issued, such as: Law no. 11 of 2008 as amended by Law no. 19/2016 on Electronic Information and Transactions (ITE LAW) also contains prohibitions and criminal threats for perpetrators who make hate speech or fake news. Criminal threats are also aimed at any person who knowingly and without rights disseminates information aimed at creating hatred or hostility for individuals and / or certain groups of society based on SARA (Article 28 paragraph (1) in conjunction with Article 45 of the ITE Law).

One thing that can be done is to use text mining technology to detect hate speech. Hate speech detection has been done a lot, and is still being done today. This research develops every year. To collect existing research and then identify, assess and interpret it, a systematic literature review (SLR) is needed [1].

SLR research was conducted to determine the performance of software engineering error prediction [2], identification and analysis related to software damage prediction systems [3], and SLR was used to support the main study selection activities using visual text mining techniques [4].

SLR research on hate speech detection with NLP has been conducted [5]. This study is aimed at new researchers, provides a brief, comprehensive and structured overview of automatic hate speech detection and outlines a systematic approach that focuses on feature extraction.

Technology is growing rapidly, hate speech detection is also evolving. Not only focusing on methods and features, this research also identifies and analyzes research trends, the data sources used in hate speech detection by gathering all the relevant evidence available to answer interesting research questions. The hope is to find out about topics that trigger a lot of hate speech. The development of hate speech detection from year to year. Besides knowing the methods and features that can be used for hate speech detection. So that it can be taken into consideration for new researchers.

SLR research collects all available relevant evidence to answer interesting research questions. Research on hate speech detection was collected from 5 digital libraries with the keywords "hate speech", "abusive" and "hate speech". The research collected is research published before the end of January 2020.

II. RESEARCH METHODOLOGY

The SLR approach is widely used for medical research as well as for software engineering research. SLR is defined as the process of gathering research evidence then identifying, assessing and interpreting it so that it provides answers to predetermined research questions [1]. This study uses a systematic approach to review the literature on hate speech detection based on the guidelines proposed by [1].

A. Research Questions

The aim of this SLR is to analyze trends, datasets, methods and features used to detect hate speech. Therefore, to keep the review focused on the objectives, the research questions outlined in Table I were made.

TABLE I. RESEARCH QUESTION

ID	Research Questions	Motivation
RQ1	What topics or case studies did the researchers choose for hate speech detection?	Identify the topic or research case study in hate speech detection
RQ2	Which data source is the most widely used to be used as a hate speech detection dataset?	Identify the dataset commonly used in hate speech detection
RQ3	What are the labels classified	Identification of labels or

Authorized licensed use limited to: Rutgers University. Downloaded on May 15,2021 at 00:11:52 UTC from IEEE Xplore. Restrictions apply.

	under hate speech detection?	output from the classification of hate speech detection
RQ4	What methods are used for hate speech detection?	Identify opportunities and trends for hate speech detection methods
RQ5	What features are used for hate speech detection?	Identification features that can be used to classify to detect hate speech
RQ6	Which method performs best when used for hate speech detection?	Identify the best method of detecting hate speech

B. *Study Selection*

Study included in this SLR is only study published in Indonesian and English, both from journal and conference papers. The study criteria that may be included in SLR are based on the inclusion and exclusion criteria which are presented in Table II.

TABLE II. INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria	Exclusion Criteria
Research in computer science, especially text mining, uses both large and small scale datasets	Research that discusses hate speech detection, non-text mining methods and frameworks.
Research that discusses or compares modeling performance in the field of hate speech detection	Research without strong validation or without experimental results on hate speech detection.
For Research that has both a conference and journal version, only the journal version will be included	Research that cannot answer the research question.
For duplicate publications of the same study, only the most complete and most recent will be included	

The research papers included in this SLR are those published before January 2020, because the search for this paper was completed on that date. Before performing a search, it is necessary to select an appropriate search database to increase the likelihood of finding highly relevant articles [3]. The list of digital databases selected for the search were: ACM Digital Library (dl.acm.org), IEEE eXplore (ieeexplore.ieee.org), ScienceDirect (sciencedirect.com), Springer (springerlink.com) and Garuda (garuda. ristekdikti.go.id). The research collected is research that discusses hate speech, so we chose the keyword "hate speech". This seems general in order to be able to find a lot of research. The next keyword is “abusive” because hate speech contains abusive sentences [6]. We added the keyword "ujaran kebencian" to find research on the detection of hate speech in Indonesia. Keyword searches are focused on titles and abstracts.

The search process and the number of studies identified at each stage are shown in Fig. 1. The research selection process was carried out in two steps: selection based on title and abstract. and full-text selection. Literature review research and other studies that do not contain experimental results are not included in the review.

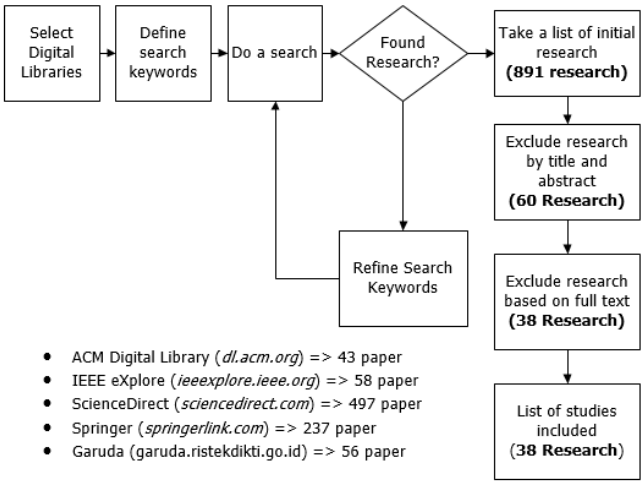


Fig. 1. Flow of research search and selection

III. RESULTS AND ANALYSIS

In this study, from 891 studies found 38 studies that fulfill the inclusion and exclusion criteria that have been designed. Groupings by year are presented to show interest in detecting hate speech. The graph of research grouping by year is shown in Fig. 2.

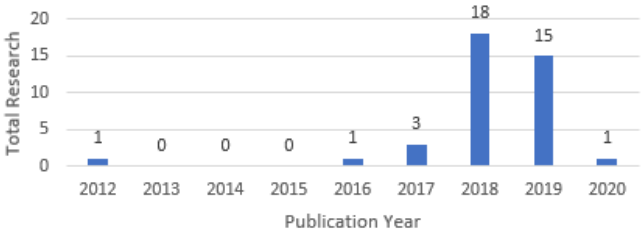


Fig. 2. Research Grouping by Year of publication

The oldest research on hate speech that researchers found was conducted in 2012 [7] when Yahoo and Facebook restricted users from posting content that is illegal, dangerous, threatening, abusive, harassing, tortuous, defamatory , vulgar, obscene, intrude on the privacy of others, hateful, pornographic and incite violence. But the prevention they do is filter user submissions that contain a list of offensive words that they specify, there is no updating of words classified as hate speech. After being stopped for 4 years, hate speech detection began again in 2016 [8] and in 2017 [9][10][11]. Research [8] is motivated by the fact that the detection of hate speech using a black list of bad words often fails because hate speech can be done more subtly and almost imperceptibly. After that, many studies have been carried out in 2018 until now. This shows that the research field on hate speech detection is still very relevant to be carried out at this time.

A. *Topics and Case Studies*

Many studies have conducted general or global case studies to detect hate speech, but there are several studies that take specific topics such as hate speech during the general election agenda, namely the 2019 Presidential Election [12] and the 2017 Pilkada [9][13][14][15]. Detection of hate speech related to political conditions was carried out by [16][17][18][19][20]. There is also research that takes the topic of hate speech related to religion [16], threats to minorities such as immigrants and homosexuals

Authorized licensed use limited to: Rutgers University. Downloaded on May 15,2021 at 00:11:52 UTC from IEEE Xplore. Restrictions apply.

[10], talks about women [21], the relationship between speech hatred with abusive words [22], detection of hate speech in comments on news pages [8] and hate speech that contains harassment [23]. The topic distribution graph is shown in Fig. 3 with the number of research units.

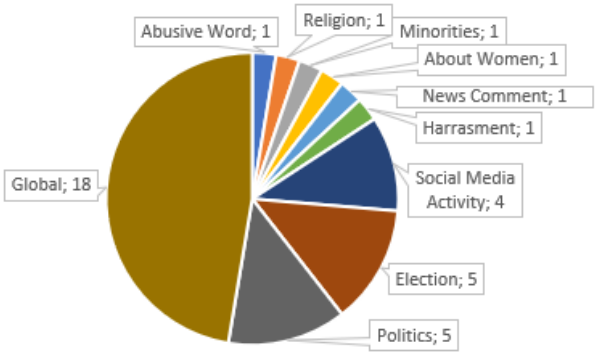


Fig. 3. Topics and Case Studies

B. Data Source

Various topics or case studies above, 65% of the studies take data sources from Twitter, then Facebook [11][12][16][24] [17][25], and other media platforms such as YouTube comments [10][26], Wikipedia talk [27][28], comments on news articles [7][29], Yahoo comment [7] [8], Instagram [30] and online today [26]. The data source is annotated to be used as a hate speech classification dataset. The portion of the data source or dataset used in hate speech detection studies is depicted in Fig. 4.

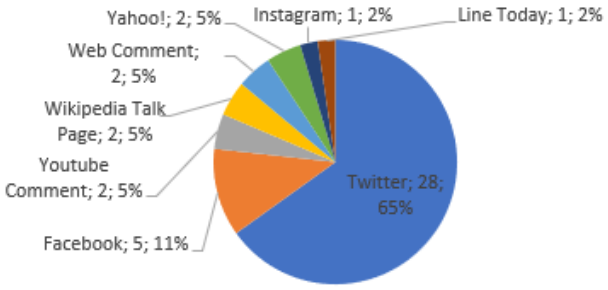


Fig. 4. Data Source

C. Class Label

Of the 38 studies we found, it turned out that they did not only classify them into 2 classes, namely hate speech (HS) and not hate speech (Non-HS). Hate speech can still be classified into HS / Offensive but Non HS / Clean [11][28][31] [32][33][34][35]; racism / sexism / clean [27][36][37]; offensive language / abusive but not offensife / Not abusive language [38]; HS-non abusive / Abusive-non HS / Abusive-HS [6]and others. Fig. 5 shows studies classifying hate speech detection into many labels.

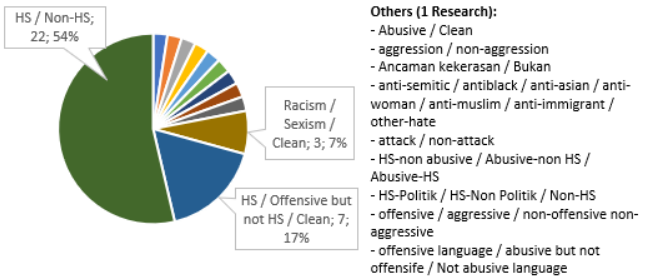


Fig. 5. Class Label

D. Classification Method

Hate speech detection is carried out by a classification method using a variety of algorithmic methods. Based on the research we found, a total of 26 classification algorithms have been used for hate speech detection as shown in Fig. 6 with the number of research units.

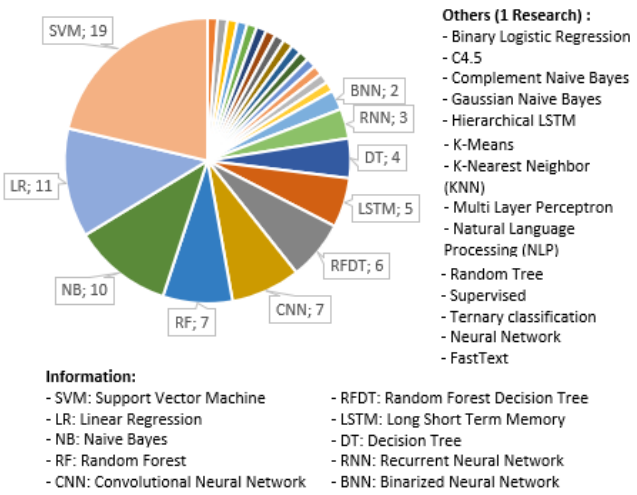


Fig. 6. Classification Method

Hate speech detection is growing, we found in research [22] where the method used is multi-label classification. This multi-label classification allows a single text to have multiple labels. In this study, hate speech is classified based on the target (individual or group), category (religion, race, physicality, gender, etc.), and level (weak, moderate, strong). The multi-label classification in this study uses an ordinary algorithm by adding a Label Power-set (LP) and Classifier Chains (CC) as data transformation methods.

E. Features

Classification performance can be influenced by the number of datasets, the features selected, the number of classes and classes that are unique to each other [36]. A bad choice to rely on only one feature to handle multiple data and inconsistencies. It is necessary to use the right combination of features. Therefore it is important to choose a set of features carefully [36]. Fig. 7 shows the features selected for the classification process for detecting hate speech in the research found.

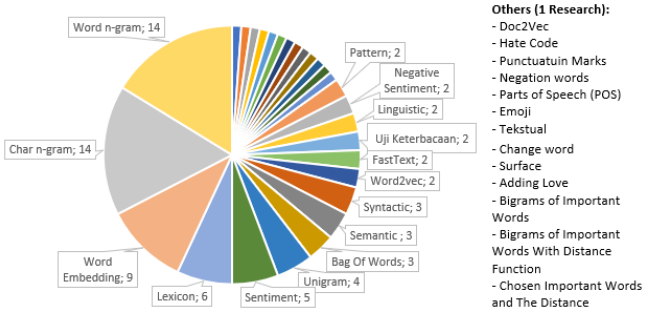


Fig. 7. Features used for hate speech detection

F. Performances

Hate speech detection performance can be evaluated using accuracy, precision, recall and f-score metrics.

Authorized licensed use limited to: Rutgers University. Downloaded on May 15,2021 at 00:11:52 UTC from IEEE Xplore. Restrictions apply.

Accuracy is the ratio of correct predictions to the total number of samples evaluated. Precision is used to measure the correctly predicted positive patterns of the total predicted patterns in the positive class. Recall is useful for measuring the fraction of positive patterns that are classified correctly. Meanwhile, F-Score or F-Measure is a measure to describe the average between recall and precision values [39].

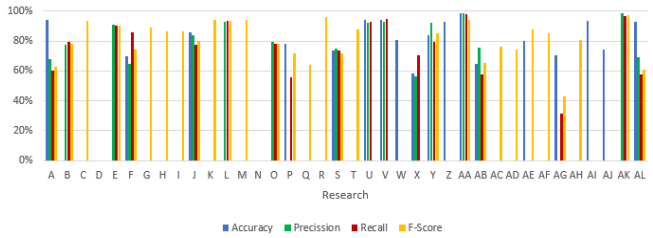


Fig. 8. Test Results of Each Study

Fig. 8 shows the results of the hate speech detection test in each of the studies found (List of research can be seen in the appendix sorted by year of publication). Not all studies use all evaluation metrics, some only calculate accuracy or F-Score. The highest accuracy is 98.71% which is obtained in research [40] which detects hate speech in Twitter social media activities. In this study, several methods were compared, namely Killer NLP by optimizing the ensemble deep learning approach (KNLPEDNN), Transformed Word Embedding Model (TWEM-MLP), NLP-SVM, and Convolution-GRUNN (CGDNN). The highest accuracy is obtained when using the KNLPEDNN method. The highest precission and recall were also obtained in research [40] with a precission value of 98.73% and a recall of 98.03%. Meanwhile, the highest F-Score is 97.40% obtained in research [36] which classifies hate speech into the labels Racism, Sexism and Clean. This study uses a data source or dataset built from Twitter and uses the Convolutional Neural Network (CNN) trained on semantic word embedding with state- of-the-art syntactic and word n-gram features.

However, this method cannot be used as a guideline or benchmark for obtaining good hate speech detection performance. This can be seen in studies that both use CNN (Uban and Liviu P Dinu, 2019) but actually get an accuracy of 64.54% with an F1-Score of 65.24%. Several studies [41][42][43] that classify HS / Non-HS and both use the SVM algorithm but have different features, have different test results (Research K, M, AG on the Fig. 8). Research [44][45] that classifies HS / Non-HS and both use the word2vec feature but the algorithm used is different, so the test results are different (Research U, V on the fig. 8).

The above confirms that the level of system performance is not only due to the use of certain algorithms but is also influenced by the number of datasets, the features selected, the number of classes and mutually exclusive classes.

IV. Conclusion

This Systematic Literature Review research aims to identify and analyze trends, datasets, methods and features used in research on hate speech detection from early to January 2020. We conducted searches in 5 digital libraries, namely ACM Digital Library (dl.acm.org), IEEE eXplore (ieeexplore.ieee.org), ScienceDirect (sciencedirect.com), Springer (springerlink.com) and Garuda (garuda.ristekdikti.go.id) with the keywords "hate speech", "abusive" and "ujaran kebencian". Of the 891 studies found, 38 studies were included in the designed inclusion and exclusion criteria. Systematic literature review is defined as the process of identifying, assessing, and interpreting all available research evidence with the aim of providing answers to research questions.

The analysis of the selected research reveals that hate speech detection research mostly takes general case studies and does not focus on specific topics. However, there are several studies that take specific topics including hate speech on the Election agenda, politics, Social Media Activities, Harassment, News Commentary, about women, minorities, religion and hate speech with and without abusive words. 65% of all research uses Twitter as a data source to be used as a dataset, 11% takes comments on Facebook posts, the rest takes comments from YouTube, Wikipedia talk, comments on web articles, yahoo, Instagram and Line Today.

This study found that hate speech was not only classified into the HS and Non-HS labels (hate speech and not hate speech) but was narrowed down into special labels. 17% of research classifies hate speech into HS / Offensive but Non HS / Clean (hate speech, speech that is offensive but not hate speech, and not both). 7% of the research classified it as Racism / Sexism / Clean. There are even those who classify it into anti-semitic / antiblack / anti-asian / anti-woman / anti-Muslim / anti-immigrant / other-hate. This shows that the detection of hate speech can be classified into labels according to the purpose of detecting hate speech.

Twenty-six different methods or algorithms have been implemented for detecting hate speech. The SVM method was most widely used in nineteen studies. Seven other algorithms that are also widely used are Linear Regression (LR), Naïve Bayes (NB), Convolutional NN (CNN), Random Forest (RF), Random Forest Decision Tree (RFDT), LSTM, and DT. Hate speech detection has increased, some are implementing multi-label classification, they are using Label Power-set (LP), and Classifier Chains (CC) as data transformation methods. Meanwhile, the most preferred features are word n-gram, char n-gram, word embedding and lexicon. In this study, it was found that it is necessary to use a combination of several appropriate features to be combined to handle various data and inconsistencies. Therefore, feature selection is one of the important things to improve hate speech detection performance.

The highest performance is obtained by the KNLPEDNN (accuracy, precision and recall) and CNN (F-Score) methods. However, the method used is not a guarantee that the classification system will get the best performance. This is evidenced by both research using the CNN algorithm resulting in poor performance. The performance of a hate speech detection classification system is also influenced by the dataset, the features selected, the number of classes and mutually exclusive classes.

References

[1] B. Kitchenham and S. Charters, "Guidelines for performing Systematic Literature Reviews in Software Enginering," pp. 1–44, 2007.

[2] T. Hall, S. Beecham, D. Bowes, D. Gray, and S. Counsell, "A systematic literature review on fault prediction performance in software engineering," IEEE Trans. Softw. Eng., vol. 38, no. 6, pp. 1276–1304, 2012.

Authorized licensed use limited to: Rutgers University. Downloaded on May 15,2021 at 00:11:52 UTC from IEEE Xplore. Restrictions apply.

[3] R. S. Wahono, "A Systematic Literature Review of Software Defect Prediction: Research Trends, Datasets, Methods and Frameworks," J. Softw. Eng., vol. 1, no. 1, pp. 1–16, 2015.

[4] K. R. Felizardo, N. Salleh, R. M. Martins, E. Mendes, S. G. Macdonell, and J. C. Maldonado, "Using visual text mining to support the study selection activity in systematic literature reviews," Int. Symp. Empir. Softw. Eng. Meas., pp. 77–86, 2011.

[5] A. Schmidt and M. Wiegand, "A Survey on Hate Speech Detection using Natural Language Processing," no. 2012, pp. 1–10, 2017.

[6] M. O. Ibrohim, M. A. Setiadi, and I. Budi, "Identification of hate speech and abusive language on Indonesian twitter using theword2vec, part of speech and emoji features," ACM Int. Conf. Proceeding Ser., 2019.

[7] W. Warner and J. Hirschberg, "Detecting hate speech on the world wide web," Proceeding LSM '12 Proc. Second Work. Lang. Soc. Media, no. Lsm, pp. 19–26, 2012.

[8] C. Nobata, J. Tetreault, A. Thomas, Y. Mehdad, and Y. Chang, "Abusive language detection in online user content," 25th Int. World Wide Web Conf. WWW 2016, pp. 145–153, 2016.

[9] I. Alfina, R. Mulia, M. I. Fanany, and Y. Ekanata, "Hate speech detection in the Indonesian language: A dataset and preliminary study," 2017 Int. Conf. Adv. Comput. Sci. Inf. Syst. ICACSIS 2017, no. October, pp. 233–237, 2017.

[10] Hugo Lewi Hammer, "Automatic Detection of Hateful Comments in Online Discussion," ICST Inst. Comput. Sci. Soc. Informatics Telecommun. Eng., vol. 188, pp. 164–173, 2017.

[11] T. Davidson, D. Warmsley, M. Macy, and I. Weber, "Automated hate speech detection and the problem of offensive language," Proc. 11th Int. Conf. Web Soc. Media, ICWSM 2017, no. Icwsm, pp. 512–515, 2017.

[12] A. S. Talita and A. Wiguna, "Implementasi Algoritma Long Short-Term Memory (LSTM) Untuk Mendeteksi Ujaran Kebencian (Hate Speech) Pada Kasus Pilpres 2019," MATRIK J. Manajemen, Tek. Inform. dan Rekayasa Komput., vol. 19, no. 1, pp. 37–44, 2019.

[13] M. M. Munir, M. A. Fauzi, and R. S. Perdana, "Implementasi Metode Backpropagation Neural Network Berbasis Lexicon Based Features dan Bag Of Words untuk Identifikasi Ujaran Kebencian pada Twitter," J. Pengemb. Teknol. Inf. dan Ilmu Komput. Univ. Brawijaya, vol. 2, no. 10, pp. 3182–3191, 2018.

[14] M. Hakiem and M. A. Fauzi, "Klasifikasi Ujaran Kebencian pada Twitter Menggunakan Metode Naïve Bayes Berbasis N-Gram Dengan Seleksi Fitur Information Gain," vol. 3, no. 3, pp. 2443–2451, 2019.

[15] A. Ramdhani, "Studi Algoritma Linear Support Vector Machine pada Deteksi Ujaran Kebencian Berbahasa Indonesia," Pros. Semin. Nas. Teknoka, vol. 3, no. 2502, p. 42, 2019.

[16] Z. Mossie and J. H. Wang, "Vulnerable community identification using hate speech detection on social media," Inf. Process. Manag., no. November 2018, p. 102087, 2019.

[17] N. Aulia and I. Budi, "Hate speech detection on Indonesian long text documents using machine learning approach," ACM Int. Conf. Proceeding Ser., pp. 164–169, 2019.

[18] N. I. Pratiwi, I. Budi, and M. A. Jiwanggi, "Hate speech identification using the hate codes for Indonesian tweets," ACM Int. Conf. Proceeding Ser., pp. 128–133, 2019.

[19] E. Sazany and I. Budi, "Deep Learning-Based Implementation of Hate Speech Identification on Texts in Indonesian: Preliminary Study," Proc. ICAITI 2018 - 1st Int. Conf. Appl. Inf. Technol. Innov. Towar. A New Paradig. Des. Assist. Technol. Smart Home Care, pp. 114–117, 2018.

[20] R. Reza El Akbar, R. N. Shofa, M. I. Paripurna, and Supratman, "The Implementation of Naïve Bayes Algorithm for Classifying Tweets Containing Hate Speech with Political Motive," ICSECC 2019 - Int. Conf. Sustain. Eng. Creat. Comput. New Idea, New Innov. Proc., no. December 2018, pp. 144–148, 2019.

[21] H. Sahi, Y. Kilic, and R. B. Saglam, "Automated Detection of Hate Speech towards Woman on Twitter," UBMK 2018 - 3rd Int. Conf. Comput. Sci. Eng., pp. 533–536, 2018.

[22] M. O. Ibrohim and I. Budi, "Multi-label Hate Speech and Abusive Language Detection in Indonesian Twitter," pp. 46–57, 2019.

[23] S. Almatarneh, P. Gamallo, F. J. R. Pena, and Alexey Alexeev, "Supervised Classifiers to Identify Hate Speech on English and Spanish Tweets," 2019.

[24] A.-S. Uban and Liviu P Dinu, "On Transfer Learning for Detecting Abusive Language Online," pp. 688–700, 2019.

[25] A. K. B. A. Putra, M. A. Fauzi, B. D. Setiawan, and E. Setiawati, "Identifikasi Ujaran Kebencian Pada Facebook Dengan Metode Ensemble Feature Dan Support Vector Machine," J. Pengemb. Teknol. Inf. Dan Ilmu Komput., vol. 2, no. 12, 2018.

[26] T. L. Sutejo and D. P. Lestari, "Indonesia Hate Speech Detection Using Deep Learning," Proc. 2018 Int. Conf. Asian Lang. Process. IALP 2018, pp. 39–43, 2019.

[27] P. Bourgonje, J. Moreno-Schneider, A. Srivastava, and G. Rehm, "Automatic classification of abusive language and personal attacks in various forms of online communication," Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), vol. 10713 LNAI, pp. 180–191, 2018.

[28] T. Gröndahl, L. Pajola, M. Juuti, M. Conti, and N. Asokan, "All You Need is 'Love': Evading Hate Speech Detection," pp. 2–12, 2018.

[29] N. D. T. Ruwandika and A. R. Weerasinghe, "Identification of hate speech in social media," 18th Int. Conf. Adv. ICT Emerg. Reg. ICTer 2018 - Proc., pp. 273–278, 2019.

[30] N. I. Pratiwi, I. Budi, and I. Alfina, "Hate speech detection on Indonesian instagram comments using FastText approach," 2018 Int. Conf. Adv. Comput. Sci. Inf. Syst. ICACSIS 2018, pp. 447–450, 2019.

[31] H. Watanabe, M. Bouazizi, and T. Ohtsuki, "Hate Speech on Twitter: A Pragmatic Approach to Collect Hateful and Offensive Expressions and Perform Hate Speech Detection," IEEE Access, vol. 6, no. c, pp. 13825–13835, 2018.

[32] R. Martins, M. Gomes, J. J. Almeida, P. Novais, and P. Henriques, "Hate speech classification in social media using emotional analysis," Proc. - 2018 Brazilian Conf. Intell. Syst. BRACIS 2018, no. October, pp. 61–66, 2018.

[33] G. Rizos, K. Hemker, and B. Schuller, "Augment to prevent: Short-text data augmentation in deep learning for hate-speech classification," Int. Conf. Inf. Knowl. Manag. Proc., pp. 991–1000, 2019.

[34] J. Dhillon, V. Gupta, R. Govil, B. Varshney, and A. Sinha, "Crowdsourcing of Hate Speech for Detecting Abusive Behavior on Social Media," 2019 Int. Conf. Signal Process. Commun. ICSC 2019, pp. 41–46, 2019.

[35] O. Oriola and E. Kotze, "Evaluating Machine Learning Techniques for Detecting Offensive and Hate Speech in South African Tweets," IEEE Access, vol. 8, pp. 21496–21509, 2020.

[36] M. Sajjad, F. Zulifqar, M. U. G. Khan, and M. Azeem, "Hate Speech Detection using Fusion Approach," 2019 Int. Conf. Appl. Eng. Math. ICAEM 2019 - Proc., pp. 251–255, 2019.

[37] G. K. Pitsilis, H. Ramampiaro, and H. Langseth, "Effective hate-speech detection in Twitter data using recurrent neural networks," Appl. Intell., vol. 48, no. 12, pp. 4730–4742, 2018.

[38] M. O. Ibrohim and I. Budi, "A Dataset and Preliminaries Study for Abusive Language Detection in Indonesian Social Media," Procedia Comput. Sci., vol. 135, pp. 222–229, 2018.

[39] Mohammad Hossin and Sulaiman M.N, "A Review on Evaluation Metrics for Data Classification Evaluations," Int. J. Data Min. Knowl. Manag. Process, vol. 5, no. 2, pp. 01–11, 2015.

[40] Z. Al-Makhadmeh and A. Tolba, "Automatic hate speech detection using killer natural language processing optimizing ensemble deep learning approach," Computing, no. 0123456789, 2019.

[41] Ziqi Zhang, D. Robinson, and Jonathan tepper, Detecting Hate Speech on Twitter Using a Convolution-GRU Based Deep Neural Network, vol. 1, no. March. Springer International Publishing, 2018.

[42] D. Robinson, Ziqi Zhang, and Jonathan tepper, "Hate Speech Detection on Twitter: Feature Engineering v.s. Feature Selection," Springer Nat. Switz. AG 2018, pp. 46–49, 2018.

[43] T. Y. S. S. Santosh and K. V. S. Aravind, "Hate speech detection in Hindi-English code-mixed social media text," ACM Int. Conf. Proceeding Ser., pp. 310–313, 2019.

[44] A. S. Saksesi, M. Nasrun, and C. Setianingsih, "Analysis Text of Hate Speech Detection Using Recurrent Neural Network," Proc. - 2018 Int. Conf. Control. Electron. Renew. Energy Commun. ICCEREC 2018, pp. 242–248, 2019.

[45] N. A. Setyadi, M. Nasrun, and C. Setianingsih, "Text Analysis for Hate Speech Detection Using Backpropagation Neural Network,"

Authorized licensed use limited to: Rutgers University. Downloaded on May 15,2021 at 00:11:52 UTC from IEEE Xplore. Restrictions apply.

APPENDIX

No	Year	Study	Label	Number of Dataset	Method	Feature
A	2012	[7]	anti-semitic / antiblack / anti-asian / anti-woman / anti-muslim / anti-immigrant / other-hate	1.000	SVM	lexicon; unigram
B	2016	[8]	Abusive / Clean	Not Listed	Supervised	char n-gram; Linguistic; Syntactic; Semantic
C	2017	[9]	HS / Non HS	1.100	NB; SVM; BLR; RFDT	BOW; Word n-gram; Char n-gram; Negatif sentiment
D	2017	[10]	Ancaman kekerasan / Bukan	24.840	LR	Bigrams of Important Words; Bigrams of Important Words with Distance Function
E	2017	[11]	HS / Offensive but not HS / Clean	Not Listed	LR; NB; DT; RF; SVM	word n-gram; Lexicon; syntactic
F	2018	[25]	HS / Non HS	Not Listed	SVM	Teksual; Lexicon
G	2018	[13]	HS / Non HS	500	BNN	BOW; Lexicon
H	2018	[15]	HS / Non HS	0	SVM	char n-gram
I	2018	[38]	offensive language / abusive but not offensife / Not abusive language	2.016	NB; SVM; RFDT	word n-gram; char n-gram
J	2018	[27]	Racism / Sexism / Clean; HS / Non-HS; aggression / non-aggression; attack / non-attack	11.304	NB; C4.5; LR	Bag of Words
K	2018	[41]	HS / Non-HS	16.914	SVM; CNN	Word embedding
L	2018	[37]	Racism / Sexism / Clean	16.000	RNN; LSTM	char n-gram
M	2018	[42]	HS / Non-HS	Not Listed	SVM; CNN	Surface; Linguistic; Sentiment
N	2018	[28]	HS / Offensive but not HS / Clean	Not Listed	LR; MLP; CNN; LSTM	word n-gram; char n-gram; change word; adding love
O	2018	[31]	HS / Offensive but not HS / Clean	24.020	Ternary classification	sentiment; Semantic; unigram; Pattern
P	2018	[21]	HS / Non-HS	1.288	SVM; DT; NB; RF; RT	word n-gram; uji keterbacaan
Q	2018	[30]	HS / Non-HS	835	FastText; LR; RFDT	word n-gram; char n-gram; fastText
R	2018	[19]	HS / Non-HS	713	LSTM	word embedding; fastText
S	2018	[29]	HS / Non-HS	1.000	NB; SVM; LR; DT; K-Means	word n-gram; uji keterbacaan; sentiment; char n-gram
T	2018	[26]	HS / Non-HS	2.273	LSTM	word n-gram; word embedding
U	2018	[44]	HS / Non-HS	1.235	RNN	word2vec
V	2018	[45]	HS / Non-HS	1.235	BNN	word2vec
W	2018	[32]	HS / Offensive but not HS / Clean	975	NB; SVM; RF	word n-gram; char n-gram; change word; adding love
X	2019	[12]	HS / Non HS	1.140	LSTM	Word embedding
Y	2019	[14]	HS / Non HS	500	NB	char n-gram
Z	2019	[16]	HS / Non-HS	14.266	RNN	word n-gram; word embedding
AA	2019	[40]	Ancaman kekerasan / Bukan	10.568	NLP; SVM; CNN	Semantic; sentiment; unigram; pattern
AB	2019	[24]	offensive / aggressive / non-offensive non-aggressive	14.100	CNN	Word embedding; Sentiment
AC	2019	[23]	HS / Non-HS	12.805	SVM; GNB; CNB; DT; KNN; RF; NN	char n-gram; doc2vec
AD	2019	[33]	HS / Offensive but not HS / Clean	25.000	CNN	Word embedding
AE	2019	[6]	HS-non abusive / Abusive-non HS / Abusive-HS	13.169	SVM; RFDT; LR	word n-gram; word embedding; pos; emoji
AF	2019	[17]	HS / Non-HS	906	SVM; LR; RFDT	word n-gram; char n-gram; lexicon
AG	2019	[43]	HS / Non-HS	3.800	SVM; RF; LSTM; Hierarchical LSTM	word n-gram; char n-gram; negation words; punctuation marks
AH	2019	[18]	HS / Non-HS	14.625	LR; SVM; NB; RFDT	word n-gram; char n-gram; hate code
AI	2019	[20]	HS-Politik / HS-Non Politik / Non-HS	20.000	NB	unigram
AJ	2019	[34]	HS / Offensive but not HS / Clean	20.000	Regression	word n-gram; char n-gram; syntactic; negative sentiment
AK	2019	[36]	Racism / Sexism / Clean	16.000	CNN; LR; RF; SVM	lexicon
AL	2020	[35]	HS / Offensive but not HS / Clean	14.896	LR; SVM; RF	word embedding

Authorized licensed use limited to: Rutgers University. Downloaded on May 15,2021 at 00:11:52 UTC from IEEE Xplore. Restrictions apply.